Seminar 3 · Instructor solutions

Demand, value functions and duality

A 90-minute route. Warm-up about 10 minutes; problem 1, including the matrix and general result, about 45; problem 2(a)–(f) about 30; final questions 5. Problem 2(g), the tangential tie, is an extension. If time is tight, let students state the repeated identities after one full derivative calculation in each problem.

Teaching focus. Work round the same four-object square twice. In problem 1 the optimiser is smooth and interior. In problem 2 utility is continuous and preferences are LNS, but convexity fails. A baseline in the first route produces two income switches at the chosen prices. At a boundary the optimal set contains two points, not a segment. Both problems use functions different from the worked Lecture 3 examples and Seminar 2. The sheet is for the teaching assistant, not a presentation.

Warm-up

The answers are **T, F, T, F, T, F**.

1. Raising the target shrinks the feasible set. Minimising the same objective over a smaller set cannot lower the minimum.
2. Every target-feasible bundle costs weakly more after a componentwise price increase. Reoptimisation can mitigate the increase, but cannot reverse this inequality. At a differentiable point $e_{p_i} = h_i \geq 0$.
3. Multiplying the objective by a positive constant preserves its minimisers; the target-feasible set does not change.
4. Convex preferences would be sufficient, but are not assumed here. Problem 2 supplies a counterexample at $p = (4, 1)$, $\bar{u} = 2$: $(1, 0)$ and $(0, 4)$ minimise expenditure, while their midpoint $(1/2, 2)$ has utility $3/2 < 2$ and does not attain the target.
5. Under the maintained conditions, $H(p, v(p, m)) = X(p, m)$ as sets. The corresponding target is the attained maximum; it is not an arbitrary number equal to income.
6. Slutsky includes the income correction: $x_{i,p_i} = h_{i,p_i} - x_i x_{i,m}$. Problem 1 gives $-6 = -4 - 2$.

1. A little of everything

For compact notation define

$$A(p) = \frac{1}{p_1} + \frac{4}{p_2}, \qquad D = p_2 + 4p_1.$$

(a) Existence, uniqueness and positive quantities

The positive-price budget is nonempty and compact. Utility is continuous, so a maximum exists. The weighted sum of square roots is strictly concave on $\mathbb{R}^2_+$, hence the maximiser on a convex budget is unique. Utility is strictly increasing in each coordinate, so the whole budget is spent.

Neither axis can contain the optimum. For example, from $(0, m/p_2)$ buy $\varepsilon > 0$ of good 1 and reduce good 2 by $p_1\varepsilon/p_2$. The gain is $\sqrt{\varepsilon}$ and the loss in the second term is $O(\varepsilon)$, since its starting quantity is positive. For small enough ε the gain dominates. The argument at the other axis uses $2\sqrt{\varepsilon}$. Do not evaluate ordinary first derivatives of the square root at zero as if they were finite.

(b) Marshallian demand and indirect utility

At the interior optimum,

$$\frac{1}{2\sqrt{x_1}} = \lambda p_1, \qquad \frac{1}{\sqrt{x_2}} = \lambda p_2.$$

Thus $x_1 = 1/(4\lambda^2 p_1^2)$ and $x_2 = 1/(\lambda^2 p_2^2)$. Exhaustion gives $A/(4\lambda^2) = m$, so $\lambda = \sqrt{A}/(2\sqrt{m}) > 0$ and

$$\boxed{x_1 = \frac{m}{p_1^2 A} = \frac{m p_2}{p_1 D}}, \qquad \boxed{x_2 = \frac{4m}{p_2^2 A} = \frac{4m p_1}{p_2 D}}.$$

Substitution gives

$$\boxed{v(p, m) = \sqrt{m A(p)}}.$$

All quantities are strictly positive for the maintained parameters.

(c) Hicksian demand and expenditure

Let $z_i = \sqrt{x_i}$. The expenditure problem becomes

$$\min_{z_1, z_2 \geq 0} \; p_1 z_1^2 + p_2 z_2^2 \quad \text{subject to} \quad z_1 + 2z_2 \geq \bar{u}.$$

The objective is strictly convex and coercive and the feasible set is closed and nonempty. The target binds. With multiplier μ on $\bar{u} - z_1 - 2z_2 \leq 0$, interior stationarity gives

$$2p_1 z_1 = \mu, \qquad 2p_2 z_2 = 2\mu.$$

The resulting positive solution is $\mu = 2\bar{u}/A$, $z_1 = \bar{u}/(p_1 A)$ and $z_2 = 2\bar{u}/(p_2 A)$. It satisfies the convex programme's sufficient KKT conditions, hence is the unique minimiser. Squaring,

$$\boxed{h_1 = \frac{\bar{u}^2}{p_1^2 A^2} = \frac{\bar{u}^2 p_2^2}{D^2}}, \qquad \boxed{h_2 = \frac{4\bar{u}^2}{p_2^2 A^2} = \frac{4\bar{u}^2 p_1^2}{D^2}}.$$

Therefore

$$\boxed{e(p,\bar{u}) = \frac{\bar{u}^2}{A(p)} = \frac{\bar{u}^2 p_1 p_2}{D}}.$$

The target is positive, so both quantities and the target multiplier are positive. The multiplier also equals $e_{\bar{u}} = 2\bar{u}/A$.

An alternative global proof. Cauchy–Schwarz gives

$$(\sqrt{x_1} + 2\sqrt{x_2})^2 \le (p_1 x_1 + p_2 x_2)\left(\frac{1}{p_1} + \frac{4}{p_2}\right).$$

It bounds utility above by $\sqrt{mA}$ on the budget and bounds the cost of the target below by $\bar{u}^2/A$. The displayed choices attain equality. This proves both values directly without relying on an unverified stationary point.

(d) Roy, Shephard and the value identities

Write $c_1 = 1, c_2 = 4$, so $A_{p_i} = -c_i/p_i^2$. Then

$$v_m = \frac{A}{2v} > 0, \qquad v_{p_i} = -\frac{mc_i}{2vp_i^2}, \qquad -\frac{v_{p_i}}{v_m} = \frac{mc_i}{p_i^2 A} = x_i.$$

Similarly,

$$e_{p_i} = -\frac{\bar{u}^2 A_{p_i}}{A^2} = \frac{\bar{u}^2 c_i}{p_i^2 A^2} = h_i.$$

The duality identities are immediate:

$$e(p, v(p,m)) = \frac{mA}{A} = m, \qquad v(p, e(p,\bar{u})) = \sqrt{(\bar{u}^2/A)A} = \bar{u}.$$

The positive-target convention removes an absolute-value ambiguity in the second equality. Substituting $\bar{u}^2 = mA$ into each Hicksian formula gives $h(p, v(p,m)) = x(p,m)$.

(e) Numbers and Slutsky

At $p = (1,4)$, $m = 8$, we have $A = 2$, $D = 8$,

$$x = (4,1), \qquad \bar{u} = v = 4, \qquad e(p,4) = 8.$$

Differentiate Marshallian demand at fixed m, and Hicksian demand at fixed $\bar{u}$ *before* substituting the matching target:

$$x_{1,p_1} = -\frac{mp_2(p_2 + 8p_1)}{p_1^2 D^2} = -6, \qquad x_{1,m} = \frac{p_2}{p_1 D} = \frac{1}{2},$$

$$h_{1,p_1} = -\frac{8\bar{u}^2 p_2^2}{D^3} = -4.$$

Thus

$$\underbrace{-6}_{\text{total}} = \underbrace{-4}_{\text{substitution}} - \underbrace{4 \cdot \tfrac{1}{2}}_{\text{income correction}}.$$

Substituting $\bar{u} = v(p,m)$ into h *before differentiating* would differentiate ordinary demand and would not compute the compensated derivative.

(f) The Slutsky matrix in this example

Compute the income correction as an outer product, not componentwise along the diagonal:

$$S = D_p x + (D_m x)x^\top = \frac{8m}{p_1 p_2 D^2}\begin{pmatrix} -p_2^2 & p_1 p_2 \\ p_1 p_2 & -p_1^2 \end{pmatrix}.$$

Independently, differentiating the Hicksian demands at fixed target gives

$$D_p h = \frac{8\bar{u}^2}{D^3}\begin{pmatrix} -p_2^2 & p_1 p_2 \\ p_1 p_2 & -p_1^2 \end{pmatrix}.$$

At $\bar{u} = v(p, m)$, $\bar{u}^2 = mA = mD/(p_1 p_2)$, so the matrices agree. The displayed matrix is symmetric. Direct multiplication gives $Sp = 0$, and

$$z^\top S z = -\frac{8m}{p_1 p_2 D^2}(p_2 z_1 - p_1 z_2)^2 \le 0.$$

At $p = (1, 4)$ and $m = 8$,

$$S = \begin{pmatrix} -4 & 1 \\ 1 & -1/4 \end{pmatrix}.$$

This is negative semidefinite, with a zero price direction. Strict negative definiteness on the whole space would be the wrong prediction.

(g) From the calculation to the general result

Under the stated regularity and duality conditions, at a matching target,

$$S = D_p h = D_{pp}^2 e.$$

Shephard gives $h_i = e_{p_i}$; Slutsky identifies the derivative matrix with the corrected Marshallian derivatives. The Cauchy–Clairaut equality of mixed partials of a C^2 function gives symmetry.

To establish the curvature, fix the target-feasible set $K = \{x \ge 0 : u(x) \ge \bar{u}\}$. For positive price vectors p and q and $0 \le t \le 1$,

$$\begin{aligned} e(tp + (1-t)q, \bar{u}) &= \min_{x \in K}\{t\, p \cdot x + (1-t) q \cdot x\} \\ &\ge t \min_{x \in K} p \cdot x + (1-t) \min_{x \in K} q \cdot x. \end{aligned}$$

Thus expenditure is concave in prices; convex preferences are not required for this argument. For any direction z, the restriction $g(t) = e(p + tz, \bar{u})$ is concave near zero, hence $g''(0) = z^\top D_{pp}^2 e\, z \le 0$. This proves negative semidefiniteness.

Finally, $h(tp, \bar{u}) = h(p, \bar{u})$: a positive common price multiplier does not change the cost rankings. Differentiating in t at 1 yields $D_p h\, p = 0$, so $Sp = 0$. Summarise the theorem only after the concrete calculation: it is a property of every regular consumer problem under these conditions, not a special feature of the square-root utility.

2. Two escape routes and a head start

Let

$$L(p,m) = 1 + \frac{m}{p_1}, \qquad R(p,m) = \sqrt{\frac{m}{p_2}}.$$

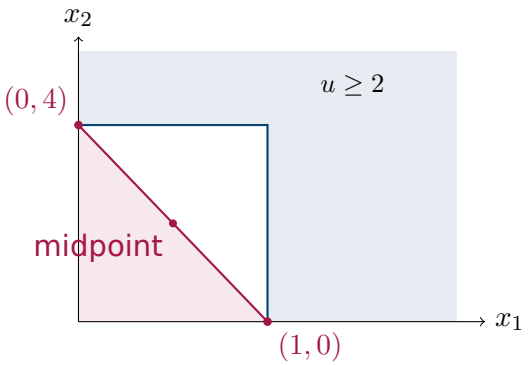
(a) Preferences, the baseline and target geometry

Utility is continuous. Increasing an active coordinate strictly improves utility arbitrarily nearby, so preferences are LNS, including at the origin. It represents complete and transitive preferences. Here $u(0) = 1$.

For $\bar{u} > 1$,

$$u(x) \geq \bar{u} \quad \Longleftrightarrow \quad x_1 \geq \bar{u} - 1 \ \textbf{or} \ x_2 \geq \bar{u}^2.$$

The set is a union. At target 2, $(1,0)$ and $(0,4)$ reach the target, but their midpoint $(1/2, 2)$ gives $\max\{3/2, \sqrt{2}\} = 3/2 < 2$. Preferences are not convex.



The red budget is $4x_1 + x_2 \leq 4$; it meets the blue target set only at the two corners. This also supplies the warm-up's counterexample.

For $\bar{u} \leq 1$, zero already attains the target; expenditure is zero and the unique cost-minimiser is zero. The main calculations use targets above the baseline. In particular, for $\bar{u} < 1$, the identity $v(p,e) = \bar{u}$ cannot hold: zero income gives utility 1, not the lower target.

(b) Marshallian demand: compare the two routes

On the budget,

$$x_1 + 1 \leq 1 + m/p_1 = L, \qquad \sqrt{x_2} \leq \sqrt{m/p_2} = R.$$

The first bound is attained at $(m/p_1, 0)$; the second component attains R at $(0, m/p_2)$. If $R < 1$, that corner's actual utility is 1, and it loses to $L > 1$ anyway. Therefore

$$\boxed{v(p,m) = \max\{L(p,m), R(p,m)\}}.$$

If both quantities are positive, both component bounds are strict, so the bundle is strictly worse than the largest corner value. There are no other maximisers. For all positive prices and income,

$$\boxed{X(p,m) = \begin{cases} \{(m/p_1, 0)\}, & L > R, \\ \{(m/p_1, 0), (0, m/p_2)\}, & L = R, \\ \{(0, m/p_2)\}, & L < R. \end{cases}}$$

At $p = (9/2, 1)$, let $t = \sqrt{m}$. Then

$$L - R = 1 + \frac{2t^2}{9} - t = \frac{2}{9}(t - 3/2)(t - 3).$$

Thus

Income	optimal route	v
$0 < m < 9/4$	good 1	$1 + 2m/9$
$9/4 < m < 9$	good 2	$\sqrt{m}$
$m > 9$	good 1	$1 + 2m/9$

At $m = 9/4$, the two corners $(1/2, 0)$ and $(0, 9/4)$ tie at utility $3/2$. At $m = 9$, $(2, 0)$ and $(0, 9)$ tie at utility 3. The baseline gives route 1 an initial advantage, the square-root route wins at intermediate incomes, and the linear route eventually wins again.

(c) Hicksian demand: compare the two costs

For $\bar{u} > 1$, the cheapest point in the first target branch is $(\bar{u} - 1, 0)$ at cost $C_1 = p_1(\bar{u} - 1)$; in the second it is $(0, \bar{u}^2)$ at cost $C_2 = p_2\bar{u}^2$. Hence

$$\boxed{e(p, \bar{u}) = \min\{C_1, C_2\}}, \qquad \boxed{H(p, \bar{u}) = \begin{cases} \{(\bar{u} - 1, 0)\}, & C_1 < C_2, \\ \{(\bar{u} - 1, 0), (0, \bar{u}^2)\}, & C_1 = C_2, \\ \{(0, \bar{u}^2)\}, & C_1 > C_2. \end{cases}}$$

Each branch has only its corner as the cheapest point at positive prices, so the tie adds exactly two bundles, not their convex combinations.

At $p = (9/2, 1)$,

$$C_1 - C_2 = \frac{9}{2}(\bar{u} - 1) - \bar{u}^2 = -(\bar{u} - 3/2)(\bar{u} - 3).$$

For $1 < \bar{u} < 3/2$ and for $\bar{u} > 3$, choose good 1. For $3/2 < \bar{u} < 3$, choose good 2. At each equality choose either corner. The expenditure values at the two switching targets are $9/4$ and 9. For all targets $\bar{u} \leq 1$, $e = 0$ and $H = \{(0, 0)\}$.

(d) Roy and Shephard within strict regimes

In a strict first-route regime, $v = 1 + m/p_1$ and

$$v_m = 1/p_1, \qquad v_{p_1} = -m/p_1^2, \qquad v_{p_2} = 0,$$

so Roy gives $(m/p_1, 0)$. In a strict second-route regime, $v = \sqrt{m/p_2}$,

$$v_m = \frac{1}{2\sqrt{mp_2}}, \quad v_{p_1} = 0, \quad v_{p_2} = -\frac{\sqrt{m}}{2p_2^{3/2}},$$

and Roy gives $(0, m/p_2)$.

If $C_1 < C_2$, $e = p_1(\bar{u} - 1)$ and $\nabla_p e = (\bar{u} - 1, 0)$. If $C_2 < C_1$, $e = p_2\bar{u}^2$ and $\nabla_p e = (0, \bar{u}^2)$. Shephard recovers the corresponding Hicksian corner. At a tie the price gradients of the two value branches differ, so the full gradient needed by Roy or Shephard is unavailable.

(e) Duality and matching the switches

Set $U = v(p, m)$. Every affordable bundle has $x_1 \le m/p_1 \le U - 1$ and $x_2 \le m/p_2 \le U^2$. More directly, $U \ge L$ implies $p_1(U-1) \ge m$, and $U \ge R$ implies $p_2 U^2 \ge m$. At least one equality holds, hence

$$e(p, v(p, m)) = \min\{p_1(U-1), p_2 U^2\} = m.$$

The equality branches are exactly the winning Marshallian routes, so the demand sets match, including at ties.

Conversely, set $M = e(p, \bar{u})$. Since $M \le p_1(\bar{u} - 1)$ and $M \le p_2 \bar{u}^2$,

$$1 + M/p_1 \le \bar{u}, \qquad \sqrt{M/p_2} \le \bar{u}.$$

At least one equality holds, so $v(p, M) = \bar{u}$. The corners attaining that equality are exactly $H(p, \bar{u})$.

At the specified prices the boundaries pair as

$$m = 9/4 \iff \bar{u} = 3/2, \qquad m = 9 \iff \bar{u} = 3.$$

No convexity of preferences is required for these duality statements.

(f) The two exact boundary examples

At $p = (9/2, 1)$, $m = 9/4$ and $\bar{u} = 3/2$,

$$X(p, 9/4) = H(p, 3/2) = \{(1/2, 0), (0, 9/4)\}.$$

The midpoint $(1/4, 9/8)$ gives $\max\{5/4, 3/(2\sqrt{2})\} = 5/4 < 3/2$ and fails the target. At the other boundary,

$$X(p, 9) = H(p, 3) = \{(2, 0), (0, 9)\}.$$

The midpoint $(1, 9/2)$ gives $\max\{2, 3/\sqrt{2}\} = 3/\sqrt{2} < 3$. Nonuniqueness persists in the corresponding expenditure problem, and the midpoint is not optimal in either problem.

(g) Extension: a tangential tie

At fixed prices $(4, 1)$,

$$1 + \frac{m}{4} - \sqrt{m} = \frac{1}{4}(\sqrt{m} - 2)^2 \ge 0.$$

Thus $v((4, 1), m) = 1 + m/4$ for every $m > 0$, and $v_m = 1/4$ even at $m = 4$. At that income, however, $(1, 0)$ and $(0, 4)$ are both optimal.

There is no contradiction with Roy: differentiability in income alone is insufficient. At the tie the price gradients are $(-1/4, 0)$ on the first branch and $(0, -1)$ on the second. The full indirect utility is not differentiable in prices there. Expenditure at fixed prices is likewise smooth in the target at 2, but its two price gradients are $(1, 0)$ and $(0, 4)$. Do not claim that every nonunique optimum forces a kink in the income-only value graph: this tie is a counterexample.

General switch geometry, for the teaching assistant. Writing $r = p_1/p_2$, the tie equation in utility is $\bar{u}^2 - r\bar{u} + r = 0$. If $r < 4$ the first route always wins. At $r = 4$ there is one tangential tie at $\bar{u} = 2$, corresponding to $m = 4p_2$. If $r > 4$, there are two thresholds

$$\bar{u}_\pm = \frac{r \pm \sqrt{r^2 - 4r}}{2} > 1, \qquad m_\pm = p_2 \bar{u}_\pm^2 = p_1(\bar{u}_\pm - 1).$$

The second route wins strictly between the thresholds and the first outside. The comparison formulas in (b)–(c) cover all prices without requiring students to carry this discriminant calculation throughout the main exercise.